SCHOOL OF ENGINEERING

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

B. TECH DEGREE 2nd SEMESTER FIRST INTERNAL EXAMINATION

APRIL 2024

EE A/EC B/IT B 23-200 - 0201B : LINEAR ALGEBRA & TRANSFORM TECHNIQUES

(2023 Scheme)

TIME: 2 hours

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to:

CO1:	Solve linear system of equations and to determine eigen values and vectors of a matrix.
CO2:	Exemplify the concept of vector space and subspace.
CO3:	Determine Fourier series expansion of functions and transform.
CO4:	Solve linear differential equation and integral equation using Laplace transform.

PART A

(Answer all the questions) **(5 x 2= 10 Marks)**

	Question	Marks	BL	CO	PO
I (a)	State Cayley-Hamilton Theorem.	2	L1	1	1
(b)	Show that the following matrix is non-singular $A = \begin{bmatrix} 1 & 2 & 4 \\ 5 & 7 & 8 \\ 6 & 1 & 2 \end{bmatrix}$	2	L2	1	2
(c)	Find the characteristic polynomial of the matrix $B = \begin{bmatrix} 6 & 5 & 2 \\ 0 & 4 & 0 \\ 1 & 3 & 9 \end{bmatrix}$	2	L3	1	1
(d)	Write down the matrix of the quadratic form $3x^2 - 8xy + y^2 - 16xz + 16yz + 5z^2$	2	L1	1	2
(e)	Find the rank of the following matrix $B = \begin{bmatrix} 5 & 7 \\ 6 & 1 \end{bmatrix}$	2	L3	1	2

PART B

(Answer the following questions)

(2 X 10= 20 Marks)

	Question	Marks	BL	CO	PO
II (a)	Find the rank of the following matrix by reducing it into normal form $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$	5	L2	1	2

(b)	Prove that a matrix A and its transpose A^T have the same characteristic roots or eigen values	5	L2	1	2

OR

III (a)	Find the eigen values and the corresponding eigen vectors of the following matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$	5	L3	1	2
(b)	Verify Cayley-Hamilton Theorem for the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and find A^{-1} and A^3	5	L3	1	2
IV (a)	Find the quadratic form of the matrix $A = \begin{bmatrix} 4 & -5 & 7 \\ -5 & -6 & 8 \\ 7 & 8 & -9 \end{bmatrix}$	5	L3	1	2
	Diagonalize the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$	5	L3	1	2

OR

V (a)	Test for consistency and solve the following system of equations $x - y + 2z = 2$ $2x + y + 4z = 7$ $4x - y + z = 4$	5	L2	2	2
(b)	Reduce the quadratic form $3x_1^2 + 3x_2^2 + 3x_3^2 + 2x_1 x_2 + 2x_1x_3 - 2x_2 x_3$ to the canonical form	5	L2	2	2

Bloom's Taxonomy Levels

L1 – 12.5%   L2- 43.75%   L3- 43.75%